

Efficient Transformer Compression via Dynamic Programming-Based Mixed-Precision Bit-Width Allocation

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Abstract—Transformer models deliver strong performance in natural language processing, but their memory footprint makes deployment difficult on resource-constrained devices. This paper studies mixed-precision quantization as an optimization problem: each Transformer layer is assigned a bit-width under a model-size budget while minimizing estimated accuracy degradation. The proposed method formulates bit-width allocation as a multiple-choice knapsack problem and solves it using Dynamic Programming.

Index Terms—Dynamic Programming, Transformer Compression, Mixed-Precision Quantization, Bit-Width Allocation, Model Compression

I. INTRODUCTION

Transformer-based language models have become the dominant architecture for modern natural language processing because they can capture long-range dependencies and transfer knowledge effectively across downstream tasks. Models derived from BERT, including DistilBERT, are widely used for classification, question answering, and semantic representation tasks. However, this accuracy comes with a substantial computational cost. Even compressed variants such as DistilBERT still store millions of parameters, and the default floating-point representation typically uses 32 bits per weight. When such models are deployed on mobile devices, embedded systems, or low-cost servers, memory consumption and inference cost become practical bottlenecks rather than secondary implementation details.

Model compression addresses this limitation by reducing the storage and computation required during inference. Quantization is one of the most direct compression techniques: model weights that are originally represented as 32-bit floating-point numbers are mapped into lower-precision representations such as 8-bit, 4-bit, or 2-bit values. This approach can significantly reduce model size because the number of bits used to store each parameter decreases. For example, replacing 32-bit weights with 8-bit weights theoretically reduces the weight storage requirement by a factor of four. For Transformer models with repeated layers and large feed-forward blocks, this reduction is especially attractive.

Uniform quantization applies the same bit-width to every layer. Although simple, this strategy ignores the fact that not

all layers have the same sensitivity to precision reduction. Some Transformer layers can tolerate aggressive quantization with only small accuracy degradation, while other layers may suffer large performance drops when forced into very low precision. This observation motivates mixed-precision quantization, in which each layer is allowed to use a different bit-width. Sensitive layers can remain at higher precision, while less sensitive layers can be compressed more aggressively. The challenge is no longer merely how to quantize a layer, but how to choose the best bit-width configuration across all layers.

The bit-width allocation problem has a combinatorial structure. If a model has L layers and each layer can choose one bit-width from a set $Q = \{2, 4, 8\}$, then the search space contains $|Q|^L$ possible configurations. A six-layer DistilBERT encoder already has $3^6 = 729$ possible layer-wise assignments for the Transformer blocks alone. Larger Transformer models make exhaustive search increasingly unattractive. More importantly, the allocation decision must satisfy a global model-size budget: choosing a high bit-width for one layer consumes budget that could have been used by another layer. Therefore, locally optimal decisions do not always produce the best global configuration.

This paper formulates mixed-precision bit-width allocation as a constrained optimization problem. Each layer corresponds to a stage, each bit-width choice corresponds to a decision, the estimated size of the quantized layer corresponds to a cost, and the measured accuracy degradation caused by fake quantization corresponds to a loss. The objective is to minimize the total estimated accuracy loss while ensuring that the total model size does not exceed a predefined budget. This formulation is equivalent in structure to the multiple-choice knapsack problem, because exactly one option must be selected from each layer group.

Dynamic Programming is used to solve the allocation problem systematically. Instead of evaluating every complete configuration independently, the algorithm builds optimal partial solutions layer by layer for each possible budget value. The state $DP[i][b]$ represents the minimum estimated accuracy loss after assigning bit-widths to the first i layers using at most budget b . By reusing subproblem solutions, the method avoids

redundant computation and guarantees the best configuration under the additive loss assumption. The selected bit-width configuration is then reconstructed through backtracking from the DP table.

The experimental framing of this work uses DistilBERT as the target Transformer model and fake quantization to estimate layer-wise sensitivity. In fake quantization, floating-point weights are rounded to a limited number of representable levels and then dequantized back to floating-point values for evaluation. This procedure does not provide hardware-level acceleration by itself, but it offers a controlled way to measure the effect of reduced precision on model behavior. The resulting sensitivity table becomes the input to the Dynamic Programming allocator. The proposed method is compared against uniform quantization and a greedy allocation baseline.

The main contribution of this paper is not a new quantization kernel or a new Transformer architecture. Instead, this work demonstrates that a classical algorithmic technique, Dynamic Programming, can be used to make mixed-precision quantization decisions in a clear, deterministic, and budget-aware manner. The contributions are threefold: first, the paper formulates Transformer bit-width allocation as a multiple-choice knapsack-style optimization problem; second, it applies Dynamic Programming to recover the optimal layer-wise bit configuration under a size constraint; third, it compares the resulting allocation with uniform and greedy baselines to analyze the trade-off between compression ratio and accuracy preservation.

II. THEORETICAL FOUNDATION

This section will discuss Transformer models, DistilBERT, quantization, mixed-precision allocation, fake quantization, and Dynamic Programming.

III. METHODOLOGY

This section will present the layer-wise sensitivity measurement procedure, the Dynamic Programming recurrence, backtracking, and baseline methods.

IV. RESULTS AND DISCUSSION

This section will report the selected bit-width configurations, model-size estimates, compression ratios, and accuracy-loss comparisons.

V. CONCLUSION

This section will summarize the role of Dynamic Programming in mixed-precision Transformer compression and discuss limitations.

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